

set operation

1) union (|)

combines both sets, common elements take as once time

```
In [3]: a = { 1 , 2 , 3 , 4 }  
        b = { 3 , 4 , 5 , 6 }  
        c = { 6 , 7 , 8 }
```

```
In [4]: a | b          # 3 , 4 take as once time
```

```
Out[4]: {1, 2, 3, 4, 5, 6}
```

```
In [5]: b | c
```

```
Out[5]: {3, 4, 5, 6, 7, 8}
```

```
In [6]: c | a | b
```

```
Out[6]: {1, 2, 3, 4, 5, 6, 7, 8}
```

```
In [8]: a = { 1 , 2 , 3 , 4 }  
        b = { 3 , 4 , 5 , 6 }  
        c = { 6 , 7 , 8 }
```

```
In [9]: print(a)  
        print(b)  
        print(c)
```

```
{1, 2, 3, 4}  
{3, 4, 5, 6}  
{8, 6, 7}
```

```
In [10]: b.union(a)
```

```
Out[10]: {1, 2, 3, 4, 5, 6}
```

2) intersection (&)

only common elements

```
In [11]: a = { 1 , 2 , 3 , 4 }  
         b = { 3 , 4 , 5 , 6 }  
         c = { 6 , 7 , 8 }
```

```
In [12]: b.intersection (c)
```

```
Out[12]: {6}
```

```
In [13]: c & a
```

```
Out[13]: set()
```

```
In [21]: a & b
```

```
Out[21]: {3, 4}
```

```
In [22]: a & b & c
```

```
Out[22]: set()
```

3) difference (-)

a - b => only print set a value, remove common from both set

```
In [ ]: b - a => only print set a value, remove common from both set
```

```
In [15]: a = { 1 , 2 , 3 , 4 }  
b = { 3 , 4 , 5 , 6 }  
c = { 6 , 7 , 8 }
```

```
In [16]: a.difference(b)
```

```
Out[16]: {1, 2}
```

```
In [17]: b - c
```

```
Out[17]: {3, 4, 5}
```

```
In [19]: b - a
```

```
Out[19]: {5, 6}
```

symmetric difference (^)

all values from both set , remove common elem

```
In [20]: a = { 1 , 2 , 3 , 4 }  
b = { 3 , 4 , 5 , 6 }  
c = { 6 , 7 , 8 }
```

```
In [23]: a ^ b
```

```
Out[23]: {1, 2, 5, 6}
```

```
In [24]: a ^ c ^ b
```

```
Out[24]: {1, 2, 5, 7, 8}
```

1) update

```
In [25]: a = { 1 , 2 , 3 , 4 }  
b = { 3 , 4 , 5 , 6 }
```

```
c = { 6 , 7 , 8 }
```

```
In [26]: a.update(b)           # only a will change    (union)
          # b is as it is
```

```
In [27]: print(a)
          print(b)
```

```
{1, 2, 3, 4, 5, 6}
```

```
{3, 4, 5, 6}
```

```
In [28]: a = { 1 , 2 , 3 , 4 }
          b = { 3 , 4 , 5 , 6 }
          c = { 6 , 7 , 8 }
```

```
In [29]: b.update(c)          # only b change
```

```
In [30]: print(b)
          print(c)
```

```
{3, 4, 5, 6, 7, 8}
```

```
{8, 6, 7}
```

2) intersection update

```
In [31]: a = { 1 , 2 , 3 , 4 }
          b = { 3 , 4 , 5 , 6 }
          c = { 6 , 7 , 8 }
```

```
In [32]: a.intersection_update(b)      # only change a
```

```
In [33]: print(a)
          print(b)
```

```
{3, 4}
```

```
{3, 4, 5, 6}
```

```
In [36]: a = { 1 , 2 , 3 , 4 }
          b = { 3 , 4 , 5 , 6 }
          c = { 6 , 7 , 8 }
```

```
In [39]: c.intersection_update(a)      #only c change
```

```
In [40]: print(a)
          print(c)      # bcoz no common elem
```

```
{1, 2, 3, 4}
```

```
set()
```

3) difference update

```
In [42]: a = { 1 , 2 , 3 , 4 }
          b = { 3 , 4 , 5 , 6 }
          c = { 6 , 7 , 8 }
```

```
In [43]: a.difference_update(b)      # only a change
```

```
In [44]: print(a)
          print(b)
```

```
{1, 2}
{3, 4, 5, 6}
```

```
In [45]: a = { 1 , 2 , 3 , 4 }
          b = { 3 , 4 , 5 , 6 }
          c = { 6 , 7 , 8 }
```

```
In [46]: c.difference_update(a)
```

```
In [48]: print(a)
          print(c)
```

```
{1, 2, 3, 4}
{8, 6, 7}
```

4) symmetric difference

```
In [54]: a = { 1 , 2 , 3 , 4 }
          b = { 3 , 4 , 5 , 6 }
          c = { 6 , 7 , 8 }
```

```
In [55]: a.symmetric_difference_update(b)      # a change only
```

```
In [56]: print(a)
          print(b)      # b as it is
```

```
{1, 2, 5, 6}
{3, 4, 5, 6}
```

superset

1 set all elem are cover into second set

```
In [60]: x={1,2,3,4,5,6}
          y={1,2,3}
          z={5,6,7}
```

```
In [61]: x.issuperset(y)      # y che all elem in x
```

```
Out[61]: True
```

```
In [62]: y.issuperset(z)
```

```
Out[62]: False
```

```
In [63]: z.issuperset(x)
```

```
Out[63]: False
```

subset

1 set all elem are cover into second set

```
In [64]: x={1,2,3,4,5,6}  
        y={1,2,3}  
        z={5,6,7}
```

```
In [65]: x.issubset(y)
```

```
Out[65]: False
```

```
In [66]: y.issubset(z)
```

```
Out[66]: False
```

disjoint

no common elem

```
In [67]: x={1,2,3,4,5,6}  
        y={1,2,3}  
        z={5,6,7}
```

```
In [68]: x.isdisjoint(y)
```

```
Out[68]: False
```

```
In [69]: x.isdisjoint(z)
```

```
Out[69]: False
```

```
In [ ]:
```